# Used-car price estimation in Kazakhstan



Machine Learning Algorithms
Endterm course project

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# Problem and dataset

## How much should a used car cost?

Brand and model
Year and mileage
Engine, body, fuel, transmission

**Regression target**

Advertised asking price
in Kazakhstani tenge (KZT)

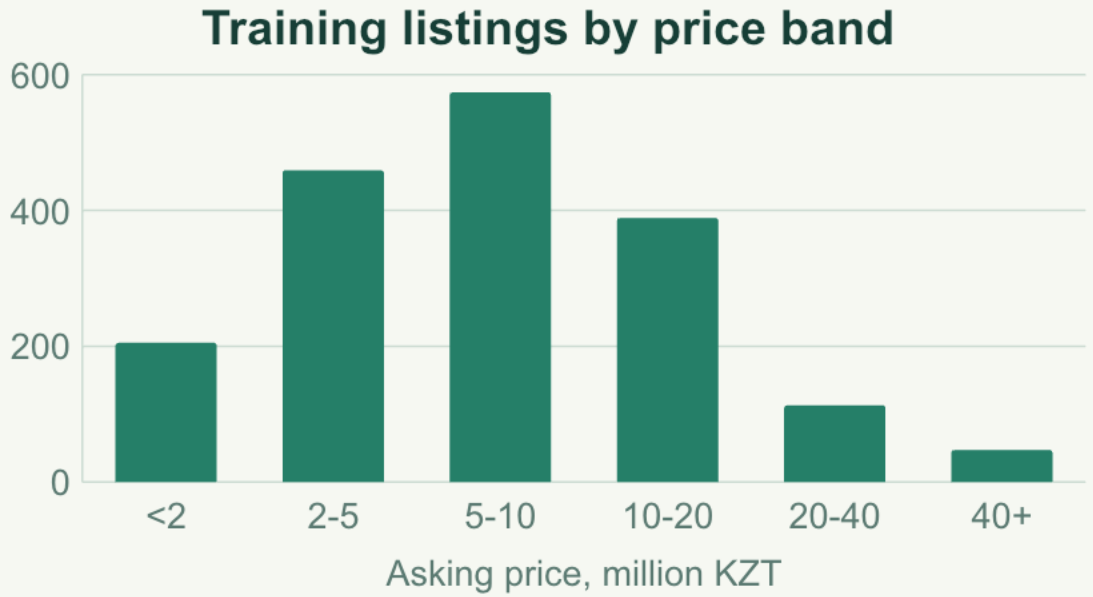
**2,553**

unique used-car listings
Kolesa-Cars-2025 on Kaggle

**Success criterion: at least 20% lower test MAE than the median baseline**

# Dataset enrichment, quality audit and EDA

| Quality audit | Count |
|---|---|
| Unique retained ads | 2,553 |
| **Enriched export** | **2,553 × 47** |
| TRAIN text with truncation signal | 1,545 / 1,787 |
| Manual-review rows | 598 |
| Recovered EV / dual fuel | 16 / 109 |




**TRAIN literal claims: colour 1,281, alloy wheels 642, sunroof 194**

Unmentioned stays NaN or unknown. Original mileage values remain unchanged.

Training mean 9.53m, median 6.70m KZT. Year ρ = 0.69, mileage ρ = -0.53.

# Validation protocol and leakage prevention

**1,787**

Training, 70%

**383**

Validation, 15%

**383**

Historical test, 15%

### Numeric features

Train medians and scaling
Mileage and engine flags
Explicit power from text

### Categorical features

Unknown is explicit
OHE thresholds 1 / 2 / 5
Native CatBoost categories

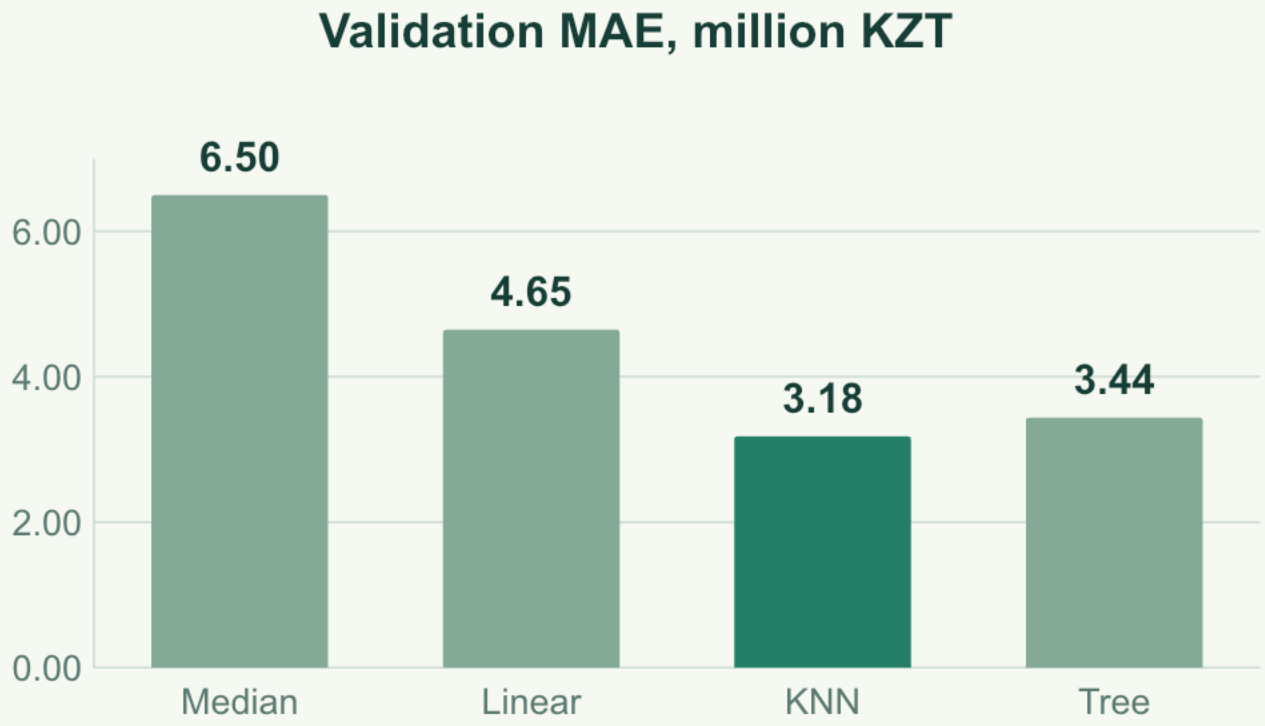
### Model decisions

Improve both validation metrics
Then minimize dual score
Freeze before test prediction

**Preprocessing fits inside training folds. Deduplicate before splitting.**

Historical errors motivated feature hypotheses. This remains a retrospective benchmark.

# Midterm baselines retained




| Model | RMSE, m KZT | $R^2$ |
|---|---|---|
| Median | 15.00 | -0.05 |
| Linear legacy | 12.25 | 0.30 |
| **KNN legacy** | **11.46** | **0.39** |
| Tree legacy | 9.78 | 0.55 |

**KNN has the lowest validation MAE among the four retained baselines**

Unchanged legacy features and configurations. The midterm comparison uses MAE.

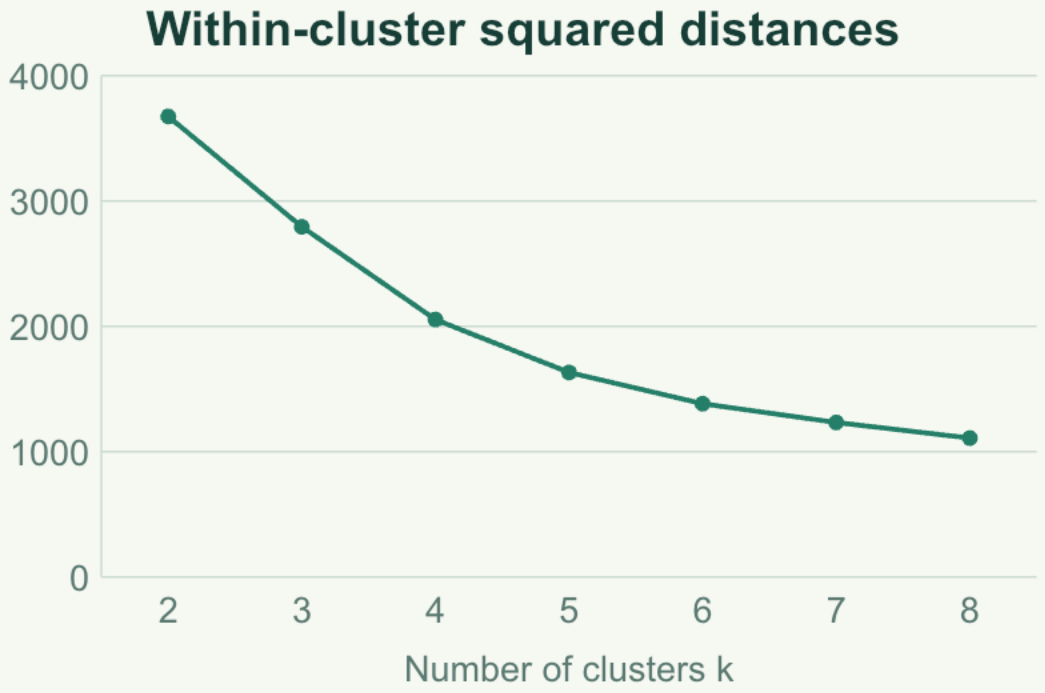
# Tuned models, CatBoost and ensemble

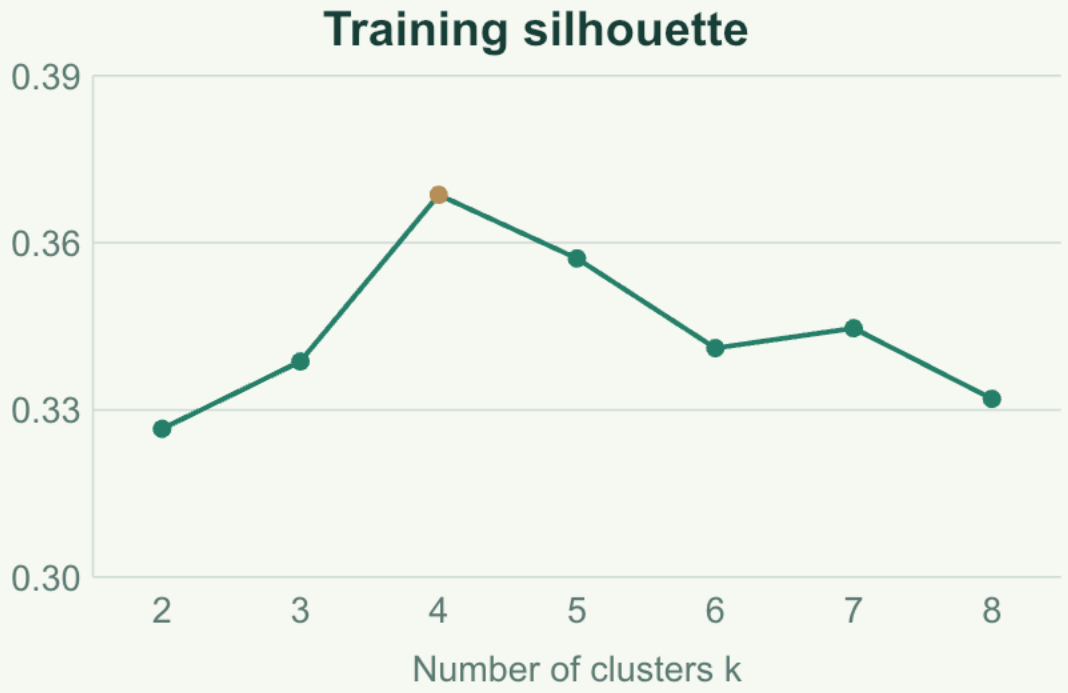
| Model | Selected configuration | CV MAE ± SD, m KZT | Val MAE, m KZT |
|---|---|---|---|
| KNN enhanced | k = 5, distance, p = 1<br>OHE min frequency = 1 | 2.62 ± 0.10 | 3.41 |
| Tree enhanced | depth = 10, min leaf = 3<br>OHE min frequency = 1 | 2.84 ± 0.14 | 3.36 |
| RF enhanced | 200 trees, max features = 0.7<br>min leaf = 1, OHE frequency = 1 | 2.17 ± 0.14 | 2.70 |
| CatBoost | depth = 8, log1p target<br>RMSE loss, 993 retained trees | 2.09 ± 0.07 | 2.69 |
| **Usage ensemble** | **25.0% new age/usage MAE pool<br>75.0% prior Robust ensemble** | **Validation only** | **2.24** |

## Only the age and usage proxies advanced from the training utility study

TRAIN utility: 15 fits. Follow-up: 15 fixed validation candidates. Other fields retained as audit data.

# K-Means cluster selection




## Selected k = 4, silhouette = 0.369

Non-electric training only (n = 1,775). Year, log mileage and engine size.

20 initializations, seed 42. Electric cars excluded from geometry; price never enters.

# K-Means cluster profiles

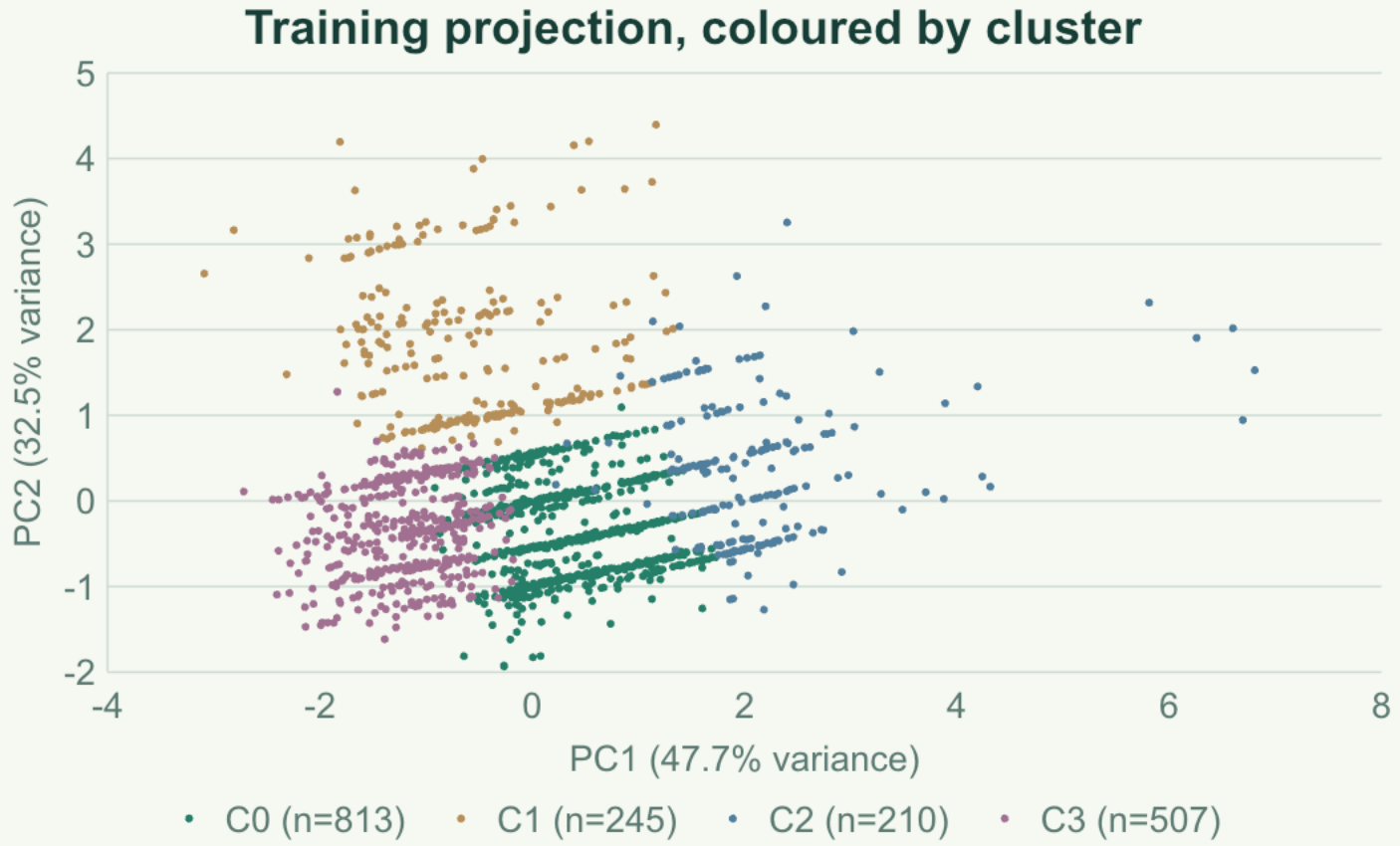
| Cluster | Listings | Year | Mileage, k km | Engine, L | Price, m KZT |
|---|---|---|---|---|---|
| C0 | 813 | 2015 | 146.0 | 2.0 | 7.69 |
| C1 | 245 | 2011 | 183.5 | 4.0 | 12.20 |
| C2 | 210 | 2023 | 23.9 | 2.0 | 14.13 |
| C3 | 507 | 1999 | 280.0 | 2.4 | 3.00 |

**C3: oldest median model year (1999)**

**C2: newest median year. C1: largest median engine (4.0 L).**

Entries are observed medians. Price is a descriptive summary added after clustering.

# PCA of the numeric vehicle features




**80.1%**

of numeric variance retained by the first two components

PC1 rises with year and falls with log mileage

PC2 mostly reflects engine size

19.9% lies outside this 2D view. PCA describes structure, not price prediction.

# Two-view ANN ensemble architecture and training

**307 / 309**
Prior / new inputs

**128**
Hidden layer, ReLU

**64**
Hidden layer, ReLU

**1**
Price per network

## Final weighted prediction

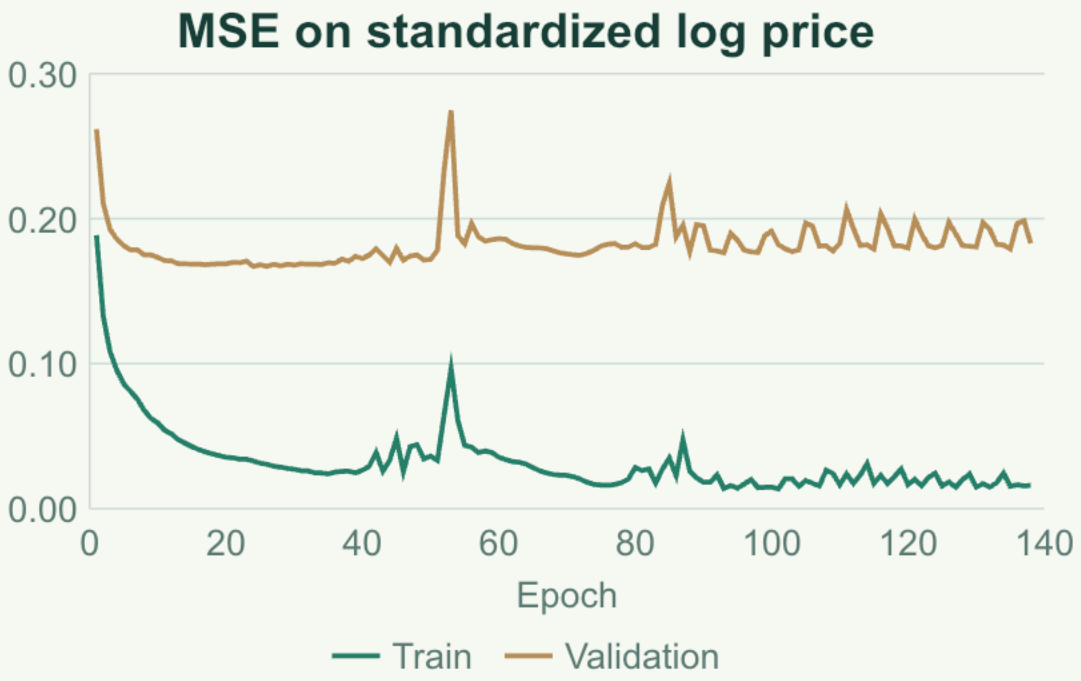
25% new MAE pool + 75% prior Robust
Five new nets: 5% each, 309 inputs
Five prior nets: 11.25% each, 307 inputs
Prior random forest: 18.75%

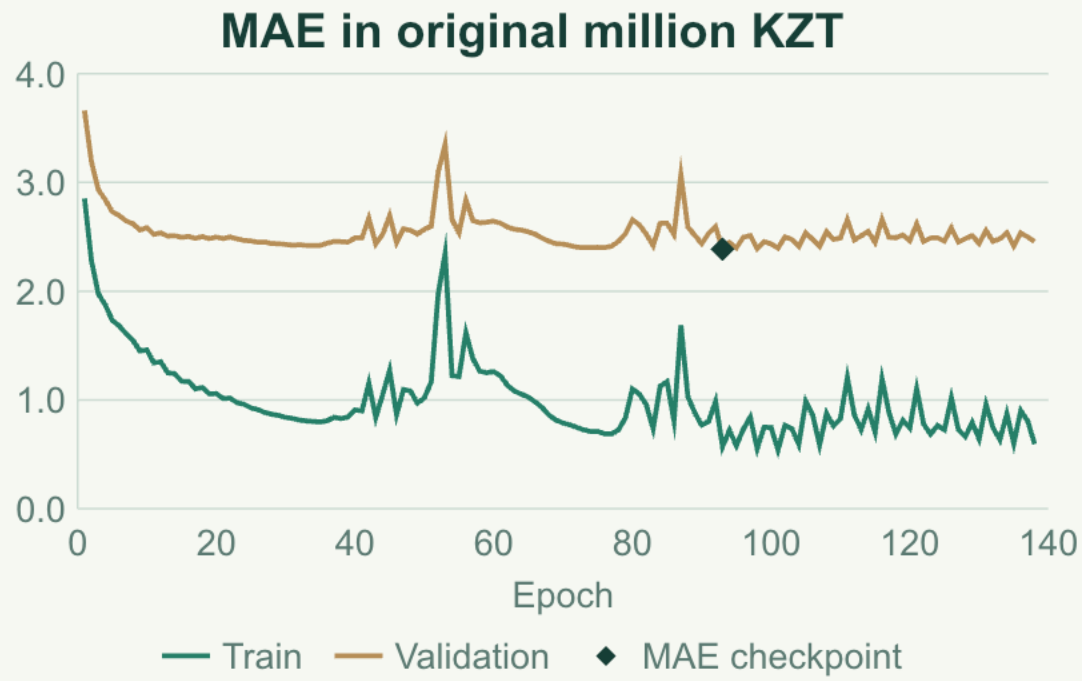
## Shared ANN training procedure

Adam 0.001, batch 64, L2 α = 0.001
Seeds 11, 23, 42, 71, 101
Train-standardized log1p(price)
New inputs: log age, log annual km proxy

128 and 64 ReLU units, linear output. Stop after 45 stale epochs, maximum 300.

# ANN pool learning curves, seed 42 example




**Seed 42: new 309-input MAE checkpoint, epoch 93 of 138**

New-pool MAE epochs: 34 / 56 / 93 / 56 / 31. Curves show seed 42 only.

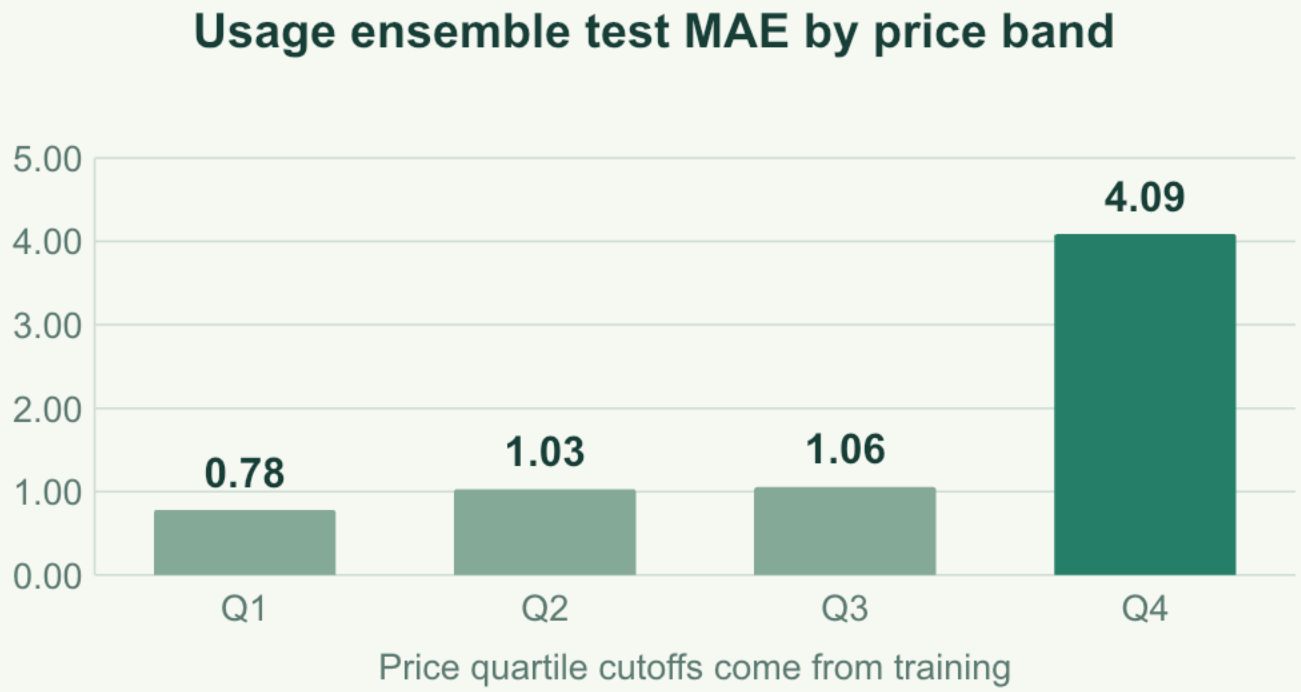
# Final model comparison

Prices are million KZT. The highlighted row wins the declared validation rule.

| Model | Val MAE | Val ±20% | Test MAE | Test ±20% | Test RMSE | Test $R^2$ |
|---|---|---|---|---|---|---|
| Median | 6.50 | 16.4% | 5.95 | 17.8% | 11.30 | -0.048 |
| Linear legacy | 4.65 | 32.9% | 3.50 | 37.6% | 6.46 | 0.657 |
| KNN legacy | 3.18 | 50.9% | 2.44 | 54.3% | 5.89 | 0.715 |
| Tree legacy | 3.44 | 46.7% | 2.46 | 48.3% | 5.48 | 0.753 |
| KNN enhanced | 3.41 | 52.7% | 2.79 | 47.8% | 6.42 | 0.662 |
| Tree enhanced | 3.36 | 45.7% | 2.53 | 52.5% | 6.21 | 0.684 |
| RF enhanced | 2.70 | 55.1% | 2.07 | 56.7% | 4.97 | 0.797 |
| ANN previous | 2.46 | 60.1% | 1.75 | 56.9% | 3.48 | 0.900 |
| ANN full | 2.45 | 61.6% | 1.85 | 55.6% | 3.88 | 0.877 |
| ANN corrected | 2.37 | 61.6% | 1.74 | 50.9% | 3.14 | 0.919 |
| CatBoost | 2.69 | 58.0% | 1.96 | 58.2% | 5.17 | 0.780 |
| Ensemble | 2.29 | 64.2% | 1.67 | 56.1% | 3.17 | 0.918 |
| ANN pool MAE | 2.27 | 65.3% | 1.65 | 55.9% | 3.13 | 0.919 |
| ANN pool dual | 2.30 | 64.8% | 1.64 | 56.7% | 3.06 | 0.923 |
| Robust ensemble | 2.25 | 66.3% | 1.65 | 57.7% | 3.25 | 0.913 |
| **Usage ensemble** | **2.24** | **67.6%** | **1.63** | **58.5%** | **3.19** | **0.917** |

Historical test, same 383 listings. Validation RMSE and $R^2$ remain in the results CSV.

# Expensive cars remain the hardest segment




**4.09m**

KZT mean absolute error in the highest price band

**Q4 mean underestimation 2.06m KZT**

Test counts: Q1 112, Q2 100, Q3 87, Q4 84

Largest error: Mercedes-Benz EQS SUV, 2022. Actual 58.00m, predicted 30.21m KZT.

# Small gains on a reused benchmark

**0.71%**

lower test MAE than the prior Robust

**95% paired intervals cross zero**

MAE reduction: −15k to +42k KZT
Within-20 gain: −0.78 to +2.61 pp
Known split, repeated validation
Small, uncertain gains. Search stopped.

**Prior Robust: 1.646m KZT**
**New blend: 1.635m KZT**

**Within ±20%: 57.7% to 58.5%**
**0.78 percentage points higher**

221 to 224 of 383 within ±20%
Median APE 16.63% to 16.61%

Median absolute error: 796k to 802k
Within ±10% and ±30%: one fewer each
RMSE 3.25m to 3.19m KZT
$R^2$ 0.913 to 0.917

# Reproducibility and contribution record

## Reproducible execution

Seed 42 and saved split IDs
Source CSV hash and pinned versions
Results tables, histories and predictions
Run: python run_project.py

Python 3.12.14
scikit-learn 1.9.1, CatBoost 1.2.10
NumPy 2.5.4, pandas 3.1.1

## Documented midterm roles

**Alisher Akhmet**

Dataset sourcing, cleaning and EDA

**Aslan Muratov**

Modelling, CV, evaluation and errors

**Endterm code and documentation are AI-assisted. Team review and defense are required.**